

Small intestine

Department of Anatomy

The abdomen (5)

First Part

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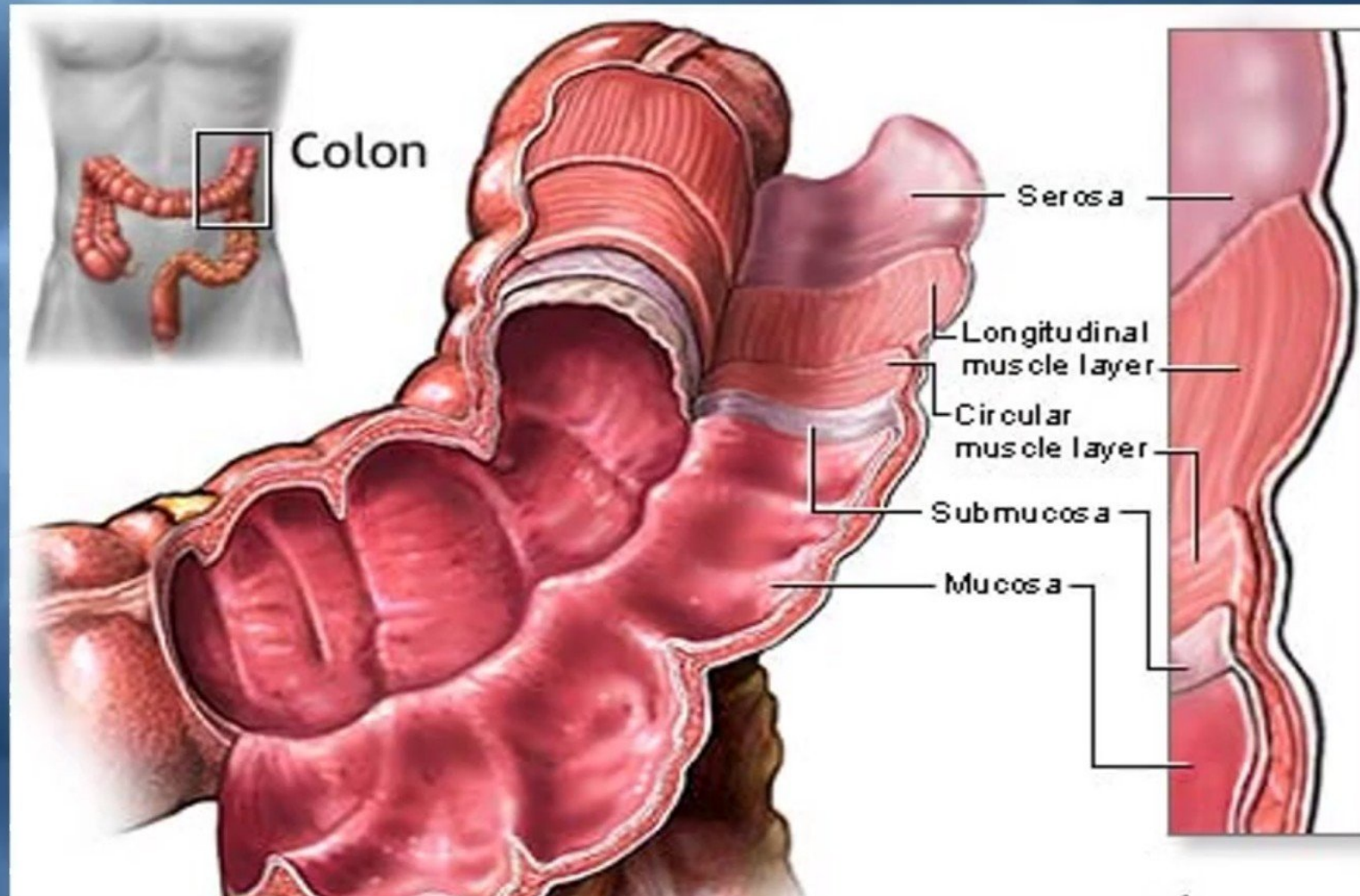
Lecture Points

- **Structure of intestine**
- **Five main functions of the bowel (intestine).**
- **Parts of the Small Intestine.**
- **The Duodenum (Location and Description)**
 - ✓ **Parts.**
 - ✓ **Length.**
 - ✓ **Direction.**
 - ✓ **Relation to Peritoneum.**
 - ✓ **First part of Duodenum.**
 - ✓ **Second part of Duodenum**
 - ✓ **Third part of Duodenum**
 - ✓ **Fourth part of Duodenum.**
 - ✓ **Nerve Supply of Duodenum**
 - ✓ **Arterial Supply of Duodenum**
 - ✓ **Veins drainage of Duodenum.**
 - ✓ **Lymph drainage of duodenum.**
- **The Jejunum & Ileum.**
- **Jejunum and Ileum (Location and Description).**
- **The root of mesentery of the small intestine**
- **The duodenojejunal flexure.**
- **The ileocecal junction.**
- **The anatomical features of Jejunum & Ileum**
- **Difference Between Jejunum & Ileum.**
- **Blood Supply of Jejunum & Ileum.**
- **Nerve Supply of Jejunum & Ileum**
- **Lymph drainage of Jejunum & Ileum**

Structure of intestine

Small & large intestine has four layers.

- 1) Peritoneal
- 2) Muscular
- 3) Submucosal
- 4) Mucosal

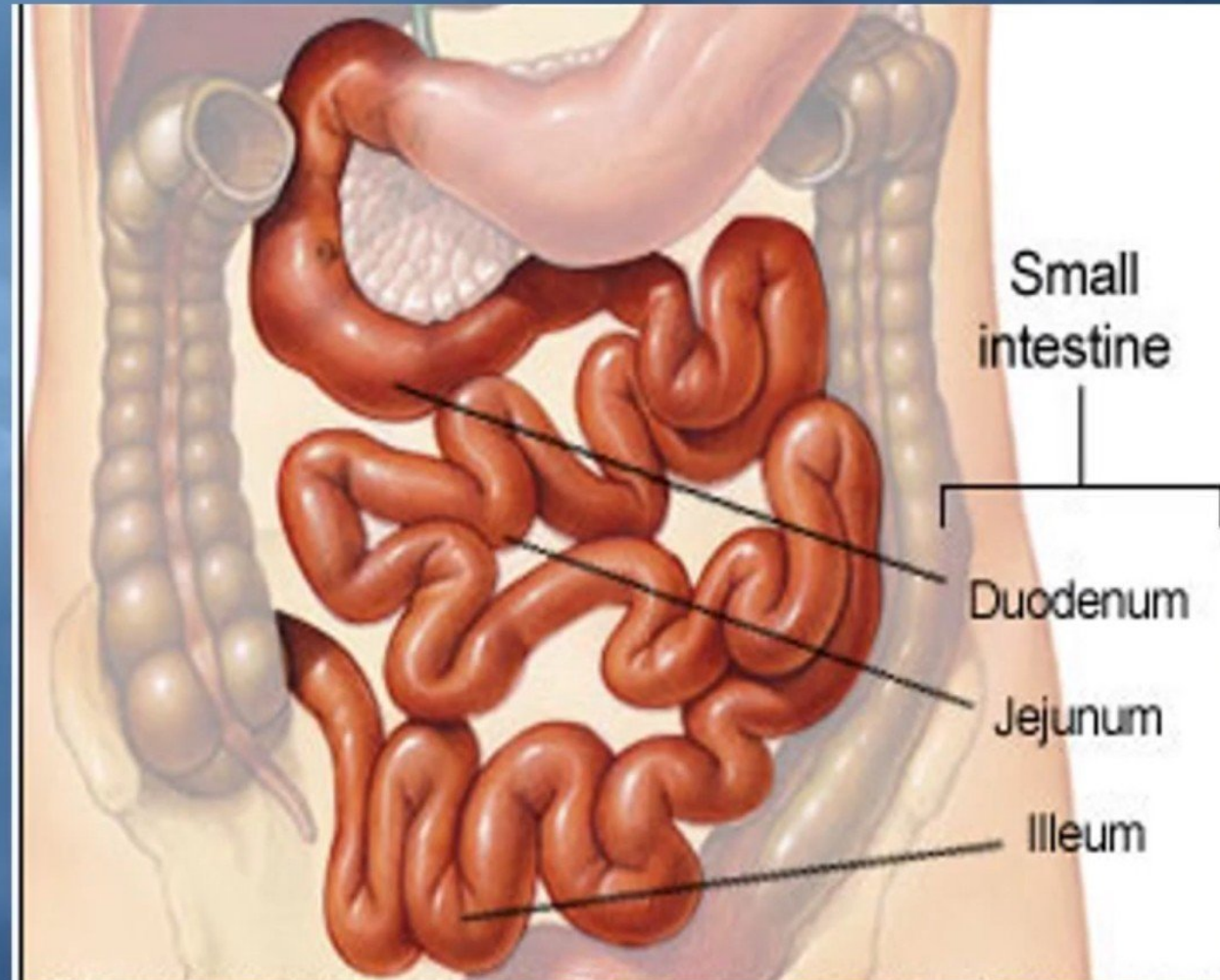


Five main functions of the bowel (intestine)

- 1) **Storage** – Unabsorbed food residue.
- 2) **Absorption** – Minerals, water, fats
- 3) **Secretion** – Mucus secreted by the colon to help lubricate the faeces
- 4) **Synthesis** – Synthesises some vitamins.
- 5) **Elimination** – Propulsion of faecal matter and absorption of fluid.

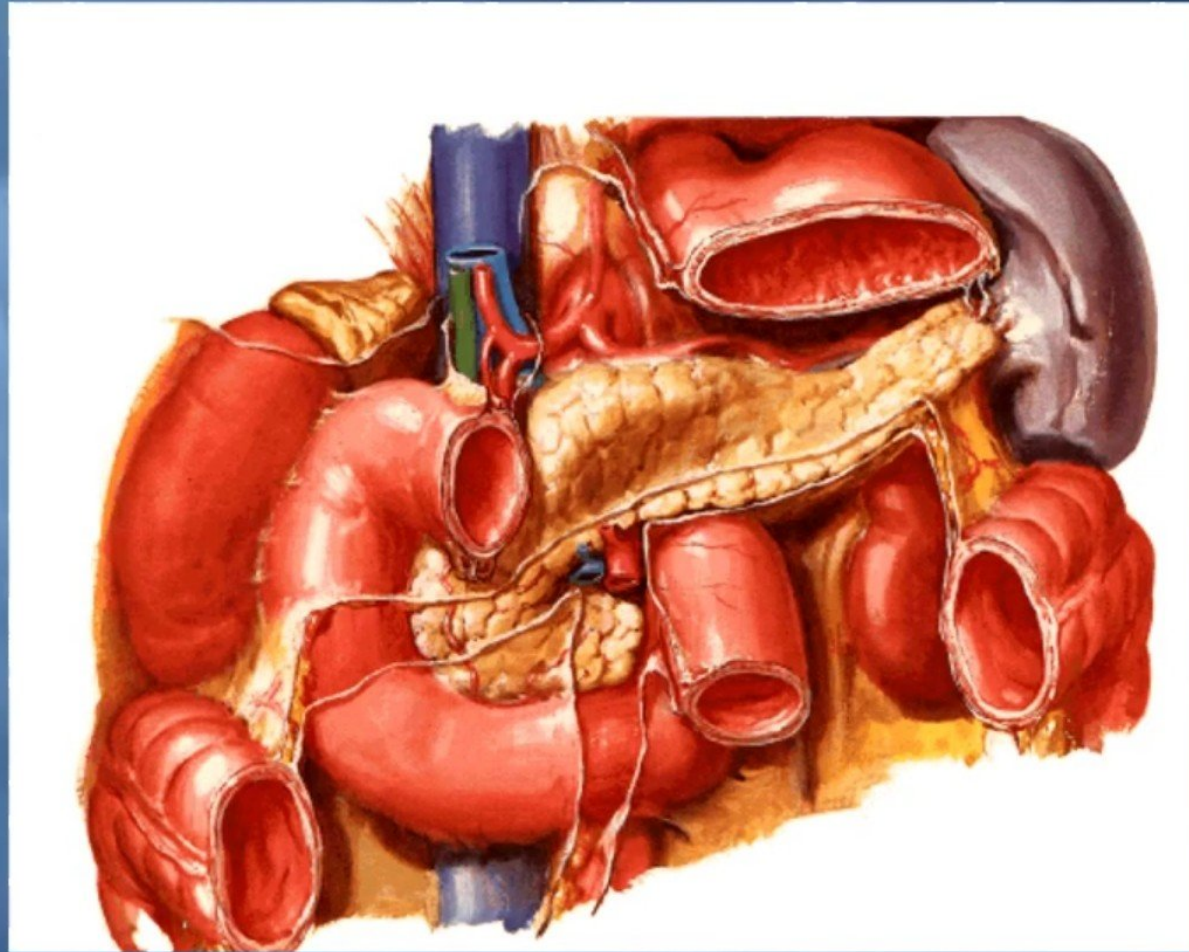
The Small Intestine

- It is **the longest part** of the alimentary canal **6 to 7 m**.
- It extends from **the pylorus** of the stomach to the **ileocecal junction**.
- It is site of most **enzymatic digestion** and **absorption**.
- It is divided into three parts:
 - 1) **Duodenum.**
 - 2) **Jejunum.**
 - 3) **Ileum.**



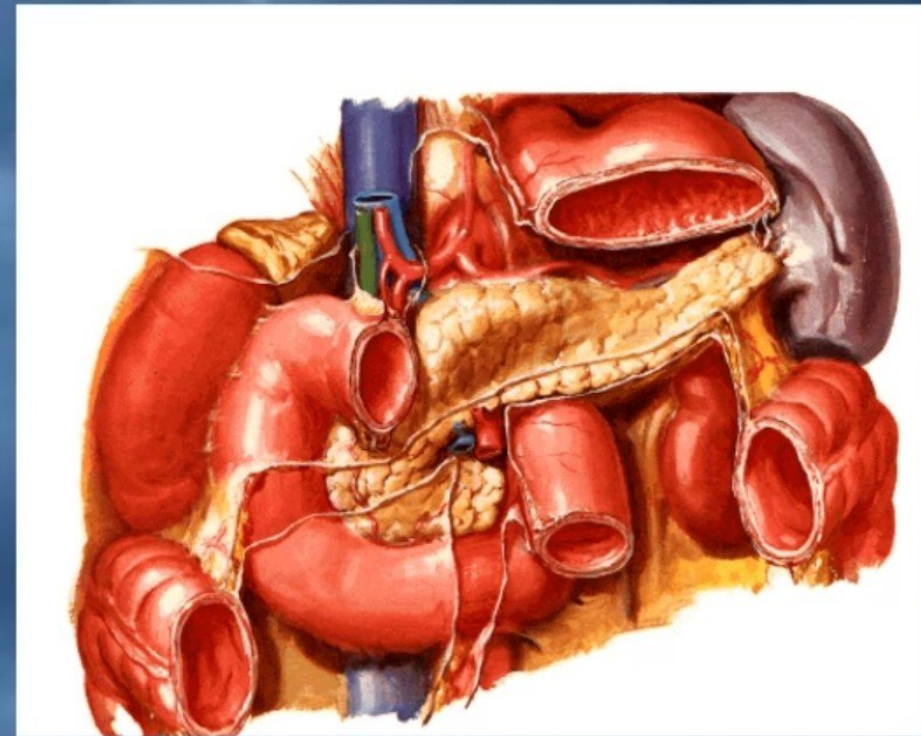
The Duodenum

- It is **the first section** of the small intestine.
- It has a **thicker layer of tissue** than the other areas of the small intestine.
- It **neutralizes stomach acids**.
- It **breaks down carbohydrates and fats**.
- It is situated in **the epigastric and umbilical regions**.
- For purposes of **description**, is **divided into four parts**.



The Duodenum (Location and Description)

- It is a **C-shaped tube**, about 10 in. (25 cm) long.
- It **joins** the **stomach** to the **jejunum**.
- It receives the **openings** of the **bile** and **pancreatic ducts**.
- The duodenum **curves around the head of the pancreas**.
- The **first inch (2.5 cm)** of the duodenum **resembles the stomach** in that it is **covered on its anterior and posterior surfaces** with peritoneum and has the **lesser omentum attached to** its **upper border** and **the greater omentum attached to** its **lower border**; the lesser sac lies behind this short segment.
- The remainder of the duodenum is **retroperitoneal**, being only **partially covered by peritoneum**.



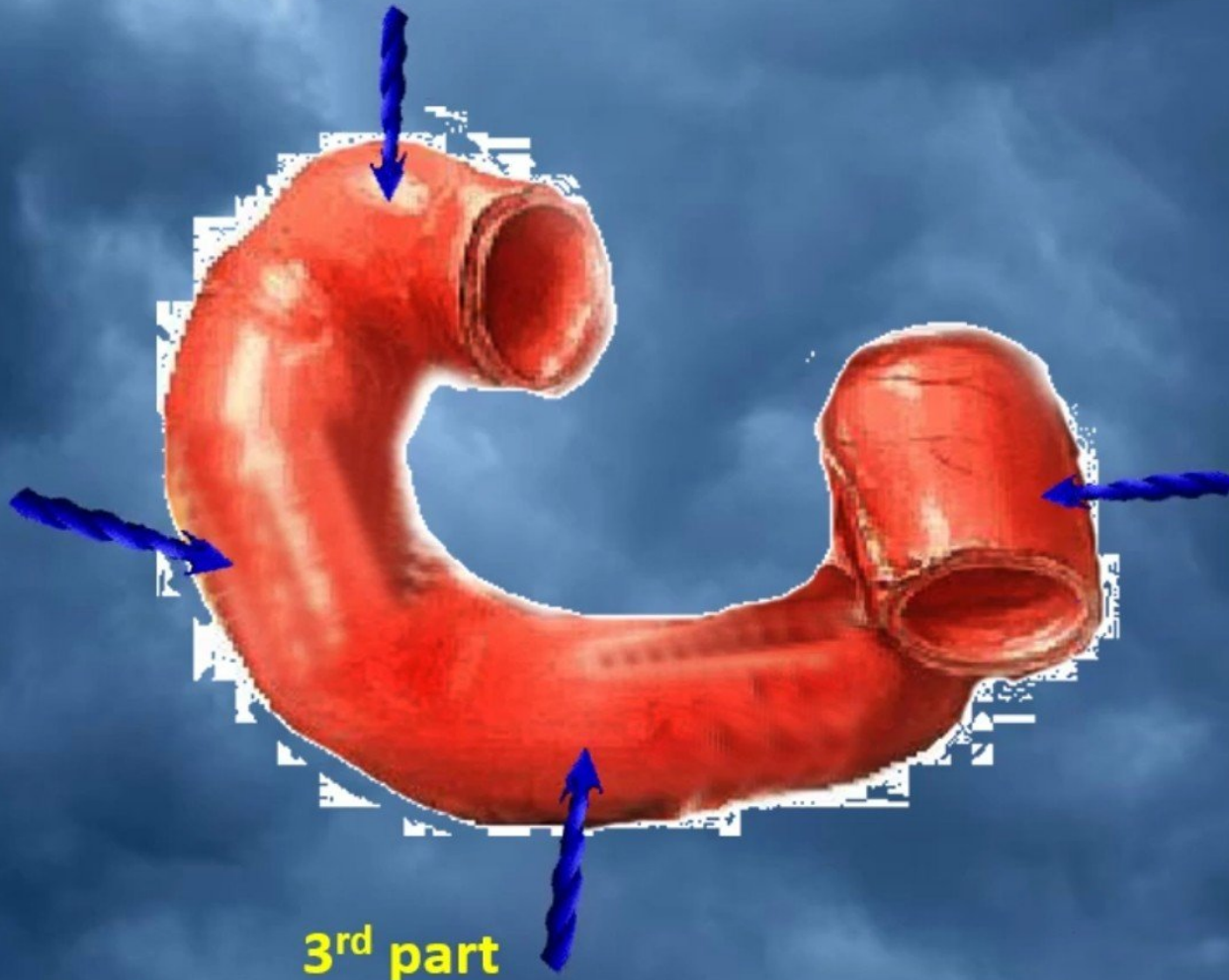
Parts of Duodenum

1st part

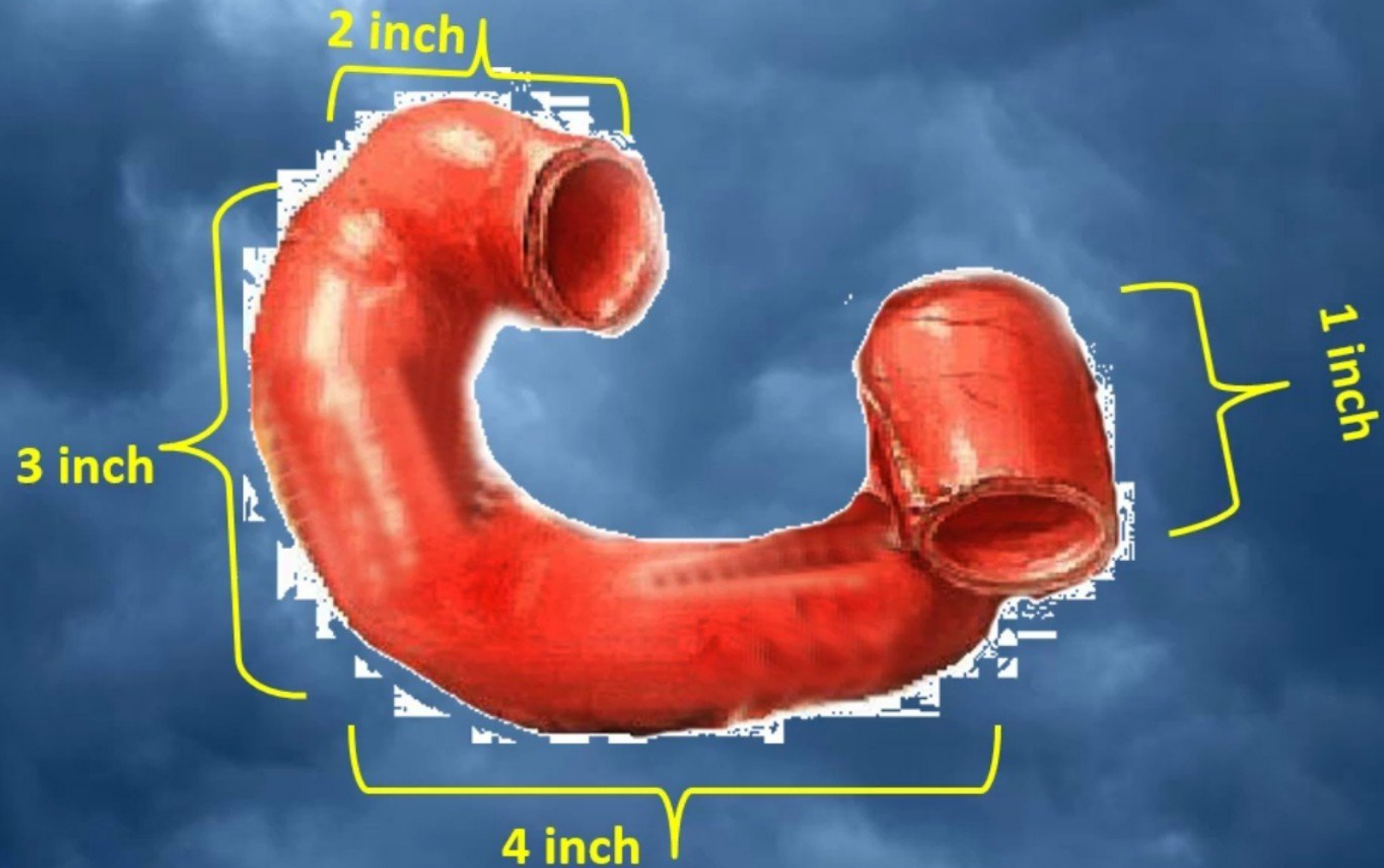
2nd part

4th part

3rd part



Length of Duodenum

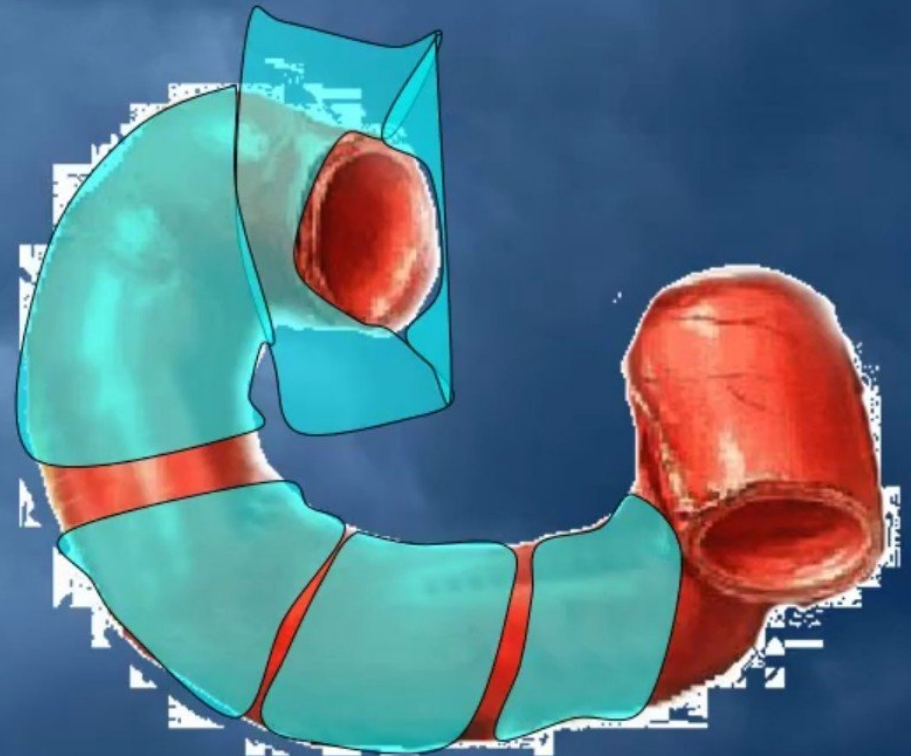


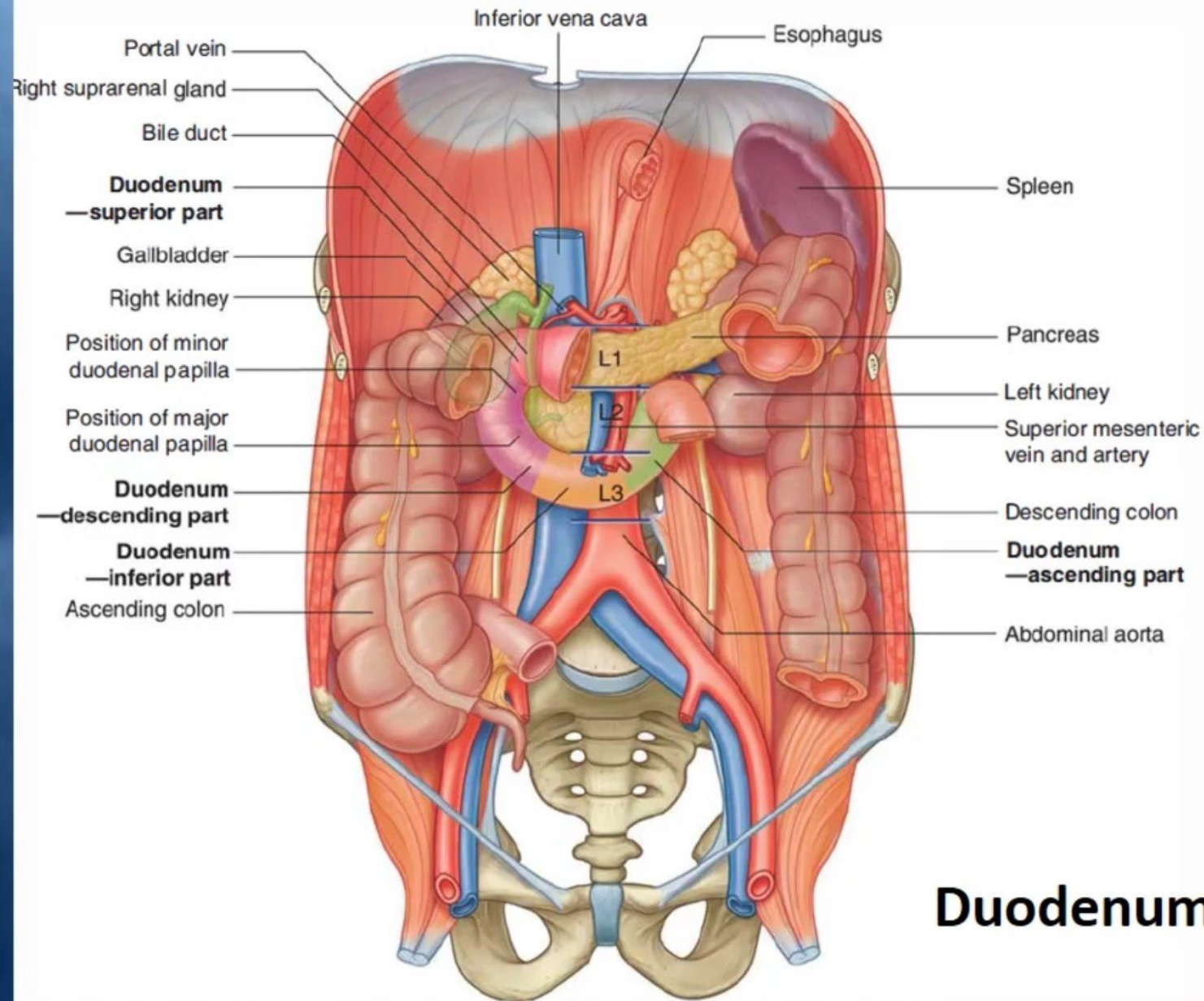
Direction of Duodenum



Relation to Peritoneum

- It is **retroperitoneal except** for the **beginning of the first part**(which is connected to the liver by the **hepatoduodenal ligament** of the lesser omentum).
- It receives blood from **the celiac (foregut) and superior mesenteric (midgut) artery**.



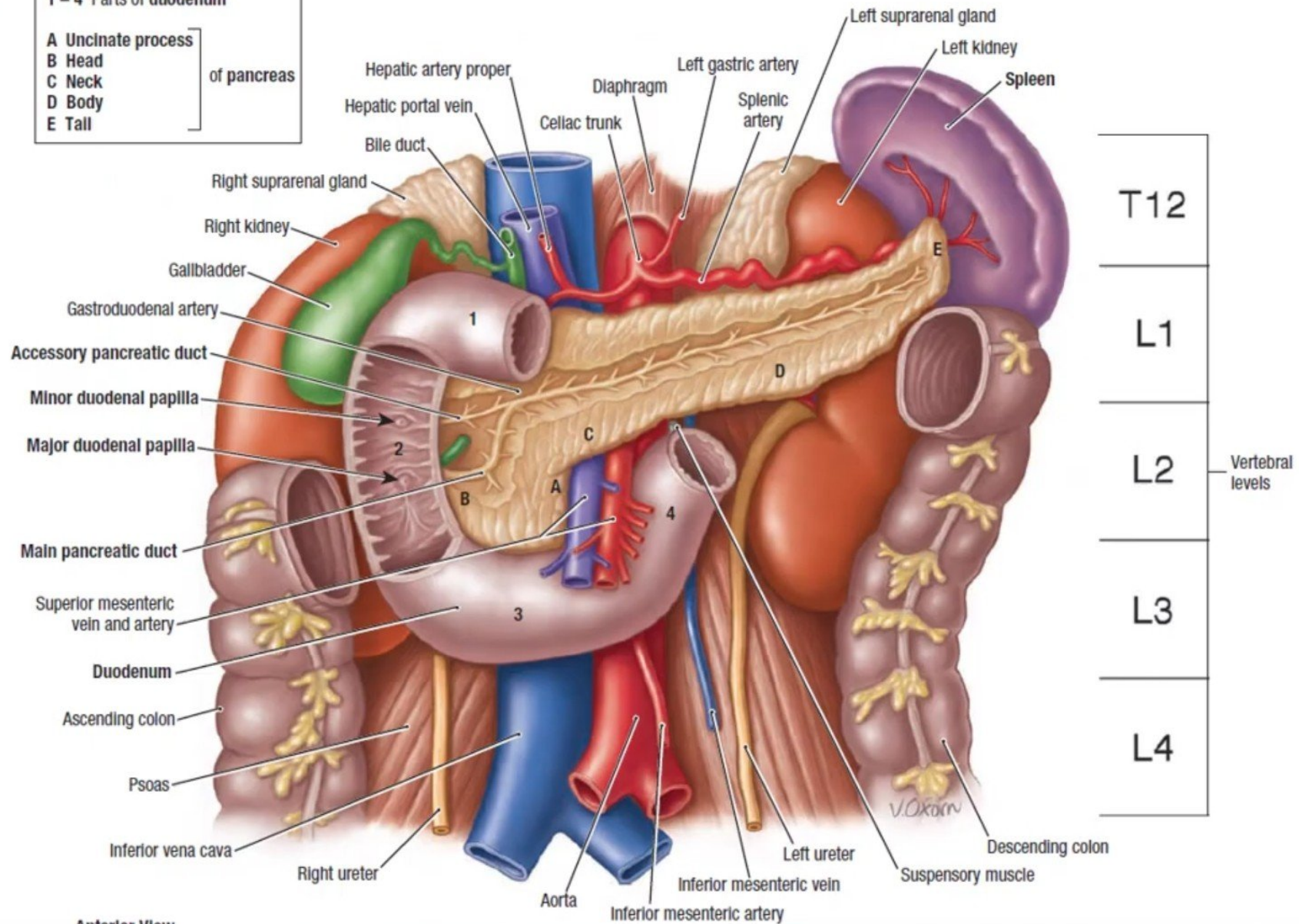


Duodenum

1 – 4 Parts of duodenum

- A Uncinate process
B Head
C Neck
D Body
E Tail

of pancreas



Will be continue...